

SHAHIL MOHAMMED

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SUMMARY

Accomplished software engineer with 3+ years of experience developing and deploying scalable applications. Skilled in full-stack development, automation, and AI integration, leveraging modern frameworks and cloud platforms to deliver impactful solutions. Proven ability to collaborate across teams and apply best practices in software engineering, security, and DevOps to drive efficiency and innovation.

PROFESSIONAL EXPERIENCE

Software Developer II, Insight Global, Austin, TX

06/2025 – Present

- Built an **Angular/TypeScript** UI with reusable components, routing, and reactive state management (RxJS), cutting layout regressions by 35% while improving feature delivery speed.
- Implemented **Java Spring Boot** REST APIs for listings, search, and details with validation and pagination, reducing average response time by 30% and improving data consistency.
- Created a **Python** solver with **AI** ranking and configurable weighting, improving match precision by 25% and enabling explainable scoring.
- Tuned **PostgreSQL** schema, indexes, and query plans for filters and keyword search, lowering query latency by 45% on large datasets.
- Upgraded authentication and role-based access control with **Spring Security**, reducing unauthorized access attempts by 40%.
- Set up an **AI-driven** code review assistant that analyzed pull requests and suggested optimizations, reducing manual review effort by 45% and accelerating development by 25%.

Software Developer, HomeTree Digital, Inc., Austin, TX

06/2024 – 06/2025

- Designed a scalable **SaaS** data visualization platform using **Laravel** and **React**, delivering real-time analytics for 50+ concurrent users.
- Built automated **Python** data pipelines to ingest and process 10,000+ stock records daily, reducing manual effort by 80%.
- Developed a secure, multi-tier e-commerce subscription system in **PHP** using **Stripe** and **WooCommerce** with role-based access control.
- Implemented production **AI agents** using **LLMs** to automate FINRA broker data processing, cutting manual workflows by 80%.
- Engineered an **AI-driven** LinkedIn outreach automation system for content scheduling, lead targeting, and engagement tracking, increasing qualified lead generation by 40%.
- Optimized **React** component rendering and state management, reducing page load times by 25% and improving overall frontend responsiveness.
- Improved platform observability by optimizing **AWS CloudWatch** metrics and alerts, achieving a 10% reduction in cloud resource wastage.

Cybersecurity Lab Support Specialist, Arizona State University, Tempe, AZ

10/2023 – 05/2024

- Maintained **Linux**-based infrastructure for cybersecurity labs, ensuring secure system configurations and reliable execution environments.
- Mentored students in **debugging**, **deploying**, and **testing** software modules, improving assignment success rates by 20%.
- Automated grading and **validation** scripts to streamline evaluation workflows, reducing manual review effort and increasing result accuracy by 25%.
- Delivered 15+ interactive online sessions covering Cryptography, Networking Protocols, and Web Security concepts, helping students improve their performance.

Software Engineer, Amrock, LLC., Detroit, MI

01/2023 – 08/2023

- Established a high-performance backend in **C#** using the **.NET** framework with efficient service-layer abstractions, achieving a 30% reduction in data retrieval latency.
- Executed and maintained **TypeScript**-based API integrations with monitoring, consistently delivering sub-100ms response times for data exchange.
- Enhanced the existing platform using **Angular.js** by improving component structure and client-side workflows, driving a 25% increase in user engagement.
- Refactored **PostgreSQL** stored procedures and query logic to improve execution plans, reducing frontend load by 15% and increasing database efficiency.
- Architected a **Kafka**-based distributed messaging pipeline with fault-tolerant producers and consumers to process 1,000+ daily events, ensuring reliable delivery and reducing message loss by 95%.
- Automated build, testing, and deployment pipelines using **Jenkins** and **Docker**, reducing release time by 40%.

Full Stack Software Engineer, Apice Tech Pvt. Ltd., New Delhi, India

05/2021 – 06/2022

- Collaborated within a 5-member engineering team to deliver a **SaaS** platform that automated key stages of the **software development life cycle (SDLC)**.
- Implemented scalable **Java Spring Boot** microservices integrated with **MySQL**, supporting 1,000 concurrent users while maintaining sub-200ms response times.
- Designed and tuned a high-performance **Oracle** SQL database through query optimization, achieving a 20% reduction in response latency.
- Engineered a scalable cloud system on **AWS EC2** using **SQS** to decouple frontend and backend, implemented custom auto-scaling to handle 1,000+ requests in under 60 seconds, and orchestrated **CI/CD** pipelines with **Docker** to deploy six production microservices reliably.
- Developed a project tracking and management UI in **React.js**, improving usability and reducing page load times by 30%.
- Validated pipeline functionality by implementing unit tests with the **JUnit** framework, achieving approximately 90% test coverage.
- Investigated production logs using **Splunk** to diagnose system issues, improving debugging efficiency and reducing incident resolution time by 20%.

EDUCATION

Arizona State University

Master of Science, Computer Science

08/2022 – 05/2024

GPA: 4.0/4.0

Manipal Institute of Technology

Bachelor of Technology, Computer and Communication Engineering

07/2018 – 07/2022

GPA: 8.83/10

TECHNICAL SKILLS

Programming & Scripting:
Web Development & Frameworks:
Database & Data Storage:
DevOps & Cloud:
Tools & Other Technologies:
AI & Automation:

Java, Python, JavaScript, PHP, Go, TypeScript, C#, C++, C, Shell Scripting
Spring Boot, React.js, Angular.js, Laravel, Node.js, Django, HTML, CSS, .NET
MySQL, SQL, PostgreSQL, MongoDB, DynamoDB, Redis, NoSQL
Amazon Web Services (AWS), Docker, Jenkins, Azure, Lambda, EC2, S3, Kubernetes
Linux/Unix, Git, GitHub, REST API, gRPC, JSON, Splunk, Jira, JUnit, WordPress
RAG, MCP, LangChain, OpenAI API, LLM Integration, AI Agents, Gen AI, Prompt Engineering

PROJECTS

AI-Powered Study Platform | Next.js, TypeScript, PostgreSQL, pgvector, Drizzle, pg-boss, OpenAI Embeddings, RAG

- Designed and implemented a PDF-to-learning pipeline using **Next.js**, **PostgreSQL/pgvector**, and **OpenAI**, cutting content creation time by 70% and generating 200+ citemed flashcards/quizzes across 10+ documents.

Material & Geometry Stress Simulation Platform | Angular, TypeScript, Node/Express, FEniCS, PostgreSQL, RAG, OpenAI

- Developed a full-stack finite-element simulation workspace using **Angular**, **Node/Express**, and **FEniCS** to automate geometry uploads, boundary-condition setup, and run comparisons, enabling interactive stress/strain analysis and mesh exports while reducing manual analysis time by 50%.

Private Communication Application | React, JavaScript, Vite, Node/Express, Socket.IO, PostgreSQL, WebRTC, Tailwind, Drizzle ORM

- Built SafeChat, a privacy-first real-time messaging platform using **React**, **Node.js**, **Socket.IO**, and **PostgreSQL**, supporting real-time chat, file sharing, and **WebRTC**-based audio/video calls with auto-expiring data, sustaining 100+ active users and enforcing a 7-day message and file retention policy.